4. Operators commuting with a projection (12 points) Let V be finite-dimensional with $\dim V = n$, and let $P \in \mathcal{L}(V)$ satisfy $P^2 = P$. Let $k = \dim \operatorname{range} P$.

a) (3) Prove that $V = \operatorname{null} P \oplus \operatorname{range} P$.

b) (4) Let $T \in \mathcal{L}(V)$. Prove that $TP = PT$ if and only if $T(\operatorname{null} P) \subseteq \operatorname{null} P$ and $T(\operatorname{range} P) \subseteq \operatorname{range} P$.

c) (3) Prove that $\mathcal{C} = \{T \in \mathcal{L}(V) : TP = PT\}$ is a subspace of $\mathcal{L}(V)$ and that $\dim \mathcal{C} = k^2 + (n-k)^2$.

d) (2) Let $P(x, y, z) = (x, y, 0)$ on $\mathbb{R}^3$. Find $\dim \mathcal{C}$, and give an operator on $\mathbb{R}^3$ that is not in $\mathcal{C}$.

Solutions

Question 1

(a) Not direct. Every $p = a(1+x^3) + b(x+x^2) \in W$ has $p(-1) = 0$ automatically, and $p(1) = 2a + 2b$. Taking $a = 1$, $b = -1$ gives

$$p = 1 - x - x^2 + x^3 = (1-x)^2(1+x) \neq 0,$$

which lies in W and vanishes at ± 1, so it lies in U. Hence $U \cap W \neq \{0\}$ and the sum is not direct. (Trap: $\dim U = 2$ and $\dim W = 2$ add to 4, but the dimension count proves nothing until the intersection is checked.)

(b) U is the null space of $p \mapsto (p(1), p(-1))$, which is surjective ($\frac{1}{2}(1+x) \mapsto (1,0)$, $\frac{1}{2}(1-x) \mapsto (0,1)$), so $\dim U = 2$. $1 + x^3$ and $x + x^2$ are independent, so $\dim W = 2$. The computation in (a) shows $U \cap W = \operatorname{span}\big(1 - x - x^2 + x^3\big)$, dimension 1. So $\dim(U + W) = 2 + 2 - 1 = 3$.

(c) $W' = \operatorname{span}(1 + x^3,\, 1)$. If $p = a(1+x^3) + b \in U$, then $p(-1) = b = 0$ and $p(1) = 2a + b = 0$, so $a = b = 0$. So $U \cap W' = \{0\}$, and $2 + 2 = 4 = \dim \mathcal{P}_3$, so $\mathcal{P}_3 = U \oplus W'$.

Question 2

(a) If $Tv = 0$ then $T^2 v = T(0) = 0$.

(b) Range: if $v \in \operatorname{null} T^2$ then $T(Tv) = 0$, so $Tv \in \operatorname{null} T$, and $Tv \in \operatorname{range} T$. Conversely if $y \in \operatorname{null} T \cap \operatorname{range} T$, write $y = Tv$; then $T^2 v = Ty = 0$, so $v \in \operatorname{null} T^2$ and $Rv = y$. Null space: $\operatorname{null} R = \operatorname{null} T \cap \operatorname{null} T^2 = \operatorname{null} T$ by (a). Rank–nullity for R on $\operatorname{null} T^2$ gives the formula.

(c) (i)$\Leftrightarrow$(ii): by (a), $\operatorname{null} T \subseteq \operatorname{null} T^2$, so they are equal exactly when the dimensions agree, which by (b) is exactly when $\dim(\operatorname{null} T \cap \operatorname{range} T) = 0$. (ii)$\Rightarrow$(iii): the sum is direct, and $\dim \operatorname{null} T + \dim \operatorname{range} T = \dim V$ by rank–nullity, so it is all of V. (iii)$\Rightarrow$(ii): a direct sum of two subspaces has zero intersection.

(d) $\operatorname{null} D = \operatorname{span}(1)$ and $\operatorname{null} D^2 = \mathcal{P}_1 \neq \operatorname{null} D$. By (c), $\mathcal{P}_3 \neq \operatorname{null} D \oplus \operatorname{range} D$. (Indeed $1 \in \operatorname{null} D \cap \operatorname{range} D$.)

Question 3

(a) $T(1) = 1$. $T(x) = (x+1) - x = 1$. $T(x^2) = (x+1)^2 - 2x^2 = -x^2 + 2x + 1$, so $T\big(x(x-1)\big) = T(x^2) - T(x) = -x^2 + 2x = -x(x-1) + x$. In the basis B:

$$\mathcal{M}(T) = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{pmatrix}.$$

(Check one column a second way: $x(x-1)$ has $p(x+1) = x^2 + x$ and $xp' = 2x^2 - x$, difference $-x^2 + 2x$.)

(b) Columns 1 and 2 are equal, so $T(x-1) = T(x) - T(1) = 0$: $\operatorname{null} T \supseteq \operatorname{span}(x-1)$. Columns 1 and 3 are independent, so $\operatorname{rank} T = 2$ and $\operatorname{null} T = \operatorname{span}(x-1)$. $\operatorname{range} T = \operatorname{span}\big(1,\ -x^2 + 2x\big)$. Not an isomorphism. (Trap: the formula looks invertible. It is not; test injectivity on a basis.)

(c) $1 + 2x - x^2 = T(x^2)$, so the solutions are $x^2 + \operatorname{null} T = \{x^2 + c(x-1) : c \in \mathbb{R}\}$.

Question 4

(a) $v = (v - Pv) + Pv$ with $P(v - Pv) = Pv - P^2v = 0$, so $V = \operatorname{null} P + \operatorname{range} P$. If $v = Pw$ and $Pv = 0$, then $v = Pw = P^2w = Pv = 0$. So the sum is direct.

(b) ($\Rightarrow$) If $Pv = 0$, then $P(Tv) = T(Pv) = 0$. If $v = Pw$, then $Tv = TPw = P(Tw) \in \operatorname{range} P$. ($\Leftarrow$) Write $v = n + r$ with $n \in \operatorname{null} P$, $r \in \operatorname{range} P$; then $Pr = r$. $TPv = Tr$. $PTv = P(Tn) + P(Tr) = 0 + Tr$, since $Tn \in \operatorname{null} P$ and $Tr \in \operatorname{range} P$ is fixed by P. So $TP = PT$.

(c) $\mathcal{C}$ contains 0, and if $S, T \in \mathcal{C}$ then $(\lambda S+T)P = \lambda SP+TP = \lambda PS+PT = P(\lambda S+T)$. Take a basis $r_1, \ldots, r_k$ of $\operatorname{range} P$ and $n_1, \ldots, n_{n-k}$ of $\operatorname{null} P$; together a basis of V by (a). By (b), $T \in \mathcal{C}$ exactly when $\mathcal{M}(T)$ in this basis is block diagonal with a $k \times k$ block and an $(n-k) \times (n-k)$ block. Since $T \mapsto \mathcal{M}(T)$ is an isomorphism, $\dim \mathcal{C} = k^2 + (n-k)^2$.

(d) $k = 2$, $n = 3$: $\dim \mathcal{C} = 4 + 1 = 5$. Not in $\mathcal{C}$: $T(x, y, z) = (z, 0, 0)$ sends $e_3 \in \operatorname{null} P$ to $e_1 \notin \operatorname{null} P$.